RICE | BUSINESS
Jones Graduate School of Business

Sourcing Data - WRDS

PhD Workshop 2026 Fall

This Workshop Session

- Objective: The session aims to introduce the WRDS platform, highlight its major features, and demonstrate how to locate the data sources available through our institution's subscription.
- After this workshop, you will be able to:
  - Navigate the WRDS website and locate the data sources available through our institutional subscription.
  - Access manuals, data dictionaries, and support resources on the WRDS platform.
  - Download data using both the web interface and SFTP.
  - Identify table names and columns of the data items.
  - Understand the data procurement process and how it works.

What is WRDS?

- Wharton Research Data Services (WRDS) is a platform that hosts datasets and provides some research resources
  - https://wrds-www.wharton.upenn.edu/pages/about/what-wrds/

- Address:
  - https://wrds-www.wharton.upenn.edu/
- Creating an Account
  - https://wrds-www.wharton.upenn.edu/register/
- Transferring an Account
  - https://wrds-www.wharton.upenn.edu/users/account-transfer/

Why use WRDS

- Most of the databases we are subscribed to are there
- Computing capabilities
- Database product documentation
- Sample code in SAS and Python for replications

Login to WRDS

- Does everyone has an account? [Create]

- Were you able to setup DUO?

- Reach out to Ben Guest and me for issues (we are the WRDS reps):
  - (email removed) – Approvals and Access
  - (email removed) - Usage

Login to WRDS

- Username and Password
  - Sometimes you get an ‘error’ just press refresh if so
- Areas of our Tour
  - Home – if you ever get ‘lost’ click on the logo
  - Data Dashboard
    - Contains all databases, segmented by the ones we have access to
    - Sometimes we have access to part of a database only, no access will be greyed out
      - Example: (i) MSCI (Climate Transition), (ii) ISS (Climate Databases)
      - What if I want a database we don’t have access to?
        - Reach out to Ben Guest for a quote
        - See if faculty are interested in purchasing – sometimes they’ll do cost-sharing

Common WRDS Databases

- **CRSP** (The Center for Research in Security Prices) – Market information (returns, prices)
- **Compustat** – Financial Statement information (Balance Sheet, Income Statement, Cash Flow)
- **Fama French Portfolios and Factors** – Kenneth French's website data (CAPM and factor models)
- **IBES** (bought First Call) – Analyst forecasts
- **TAQ** (Trade and Quote) – Intraday securities data
- **Thomson Reuters** – Mutual funds and institutional holdings
- **Execucomp** – Executive compensation from COMPUSTAT

‘Seasonal’ WRDS Databases

- Revelio Labs – Workforce database
  - Data Dictionary
  - Access until ~Feb 2027

- Ravenpack – News and their sentiment
  - Access until ~August 2027

Contributed WRDS Databases

- Contributed Data
  - Authors upload data they constructed for their papers
  - May require an underlying subscription to a product
  - Example:
    - Peters and Taylor Q

WRDS Tour

- Videos
  - https://wrds-www.wharton.upenn.edu/pages/grid-items/downloading-the-sp-500-constituents/

- Research
  - Applications
    - PEAD, SUE, DGTW, Event Study, Linking Tables

- Note: We do not have other historical index memberships [link]

WRDS Tour

- Research
  - Applications
    - Python - has some of the programs replicated in Python that were done in SAS
    - Institutional Ownership
    - Microstructure
  - Macros (SAS) - this used to be a thing, now no one cares
    - Winsorize
    - FFI
    - FM
    - ICLINK – creates a linking table between IBES and CRSP

WRDS Tour – Some caveats

- Sometimes some of the information is outdated
  - Instead of using the code ICLINK to merge IBES with CRSP there is now a linking table (BETA)
    - It can be found in linking databases under Ways to Research
  - Some of the CRSP/Compustat Linking table doesn't apply
    - https://wrds-www.wharton.upenn.edu/pages/support/manuals-and-overviews/crsp/crspcompustat-merged-ccm/wrds-overview-crspcompustat-merged-ccm/
    - https://wrds-www.wharton.upenn.edu/pages/support/support-articles/ccm/crsp-ccm-changes-2014/
    - You can see a discussion of usedflag in the CCMXPF_LINKTABLE, but that dataset is no longer updated. You should use CCMXPF_LINKHST.

WRDS Tour

- Support
  - Let’s try to find information about how to adjust prices for stock splits in CRSP
    - adjust price crsp



Support > Knowledgebase > CRSP > Stock

Adjusted Prices for Stock Splits and Dividends

Adjusted Prices for Stock Splits and Dividends

Content Body:
Detail In CRSP Stocks, returns (RET) are already adjusted -- however, prices (PRC) and shares outstanding (SHROUT) are not. The variable CFACPR (Cumulative Factor to Adjust Price) tracks the net (cumulative) level of splits over time. There is a related variable to adjust shares (CFACSHR), which is often the same as ...

Content Type: Content

  - https://wrds-www.wharton.upenn.edu/pages/support/support-articles/crsp/stock/adjusted-prices-stock-splits-and-dividends/
  - Cumulative Adjustment Factors in Stock v2

WRDS Tour – CRSP Changes

- CRSP recently changed the structure of their data
  - Old version is called SIZ, new version is called CIZ
  - Some of the code provided by WRDS already uses CIZ
  - There are many changes, for example if you want to filter by share code

- Overall changes summary by WRDS
  - https://wrds-www.wharton.upenn.edu/pages/support/manuals-and-overviews/crsp/stocks-and-indices/crsp-stock-and-indexes-version-2/crsp-ciz-faq/

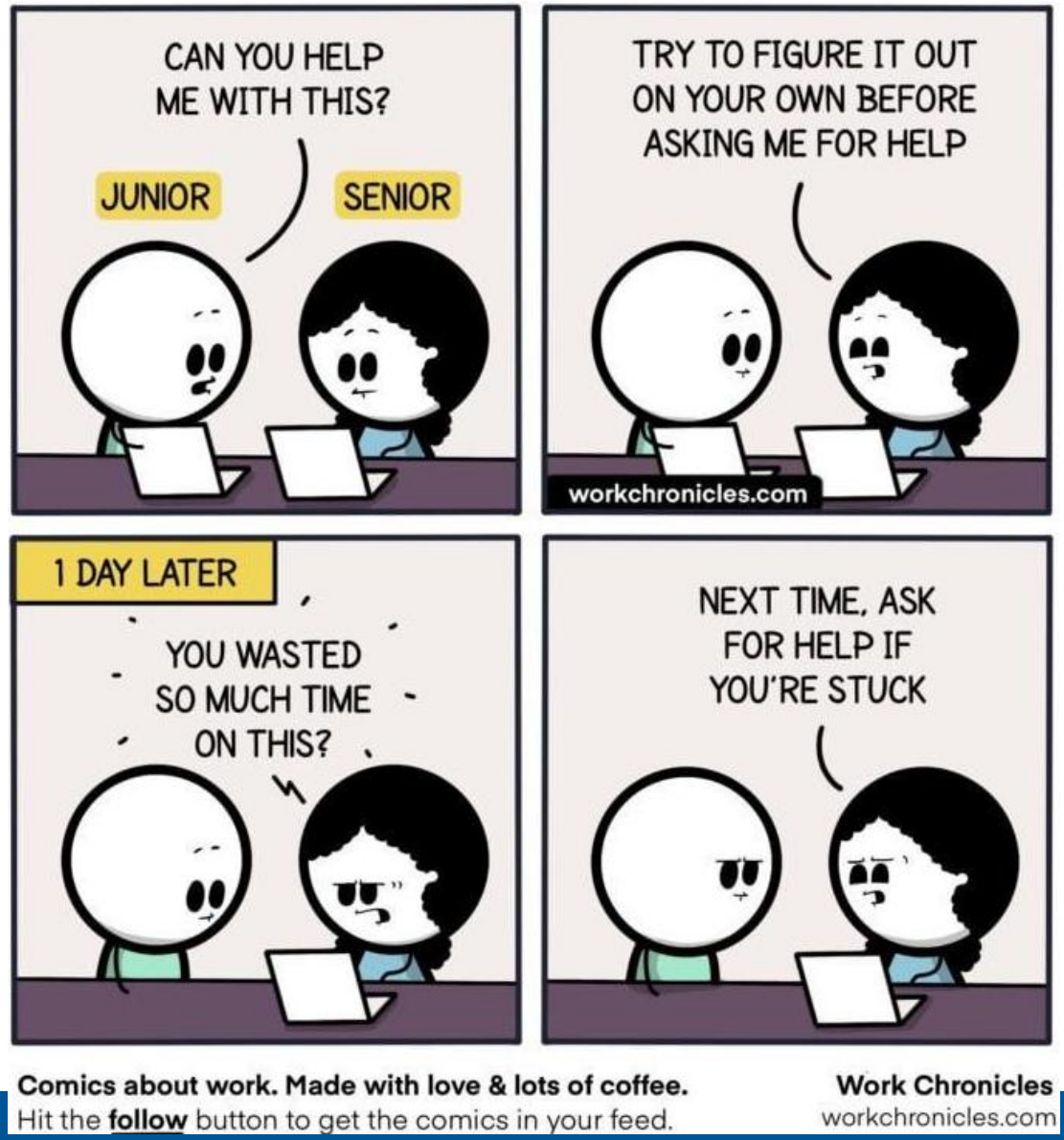
WRDS Tour – Compustat Changes

- Legacy to Xpressfeed (2008)

- Older papers will have the Legacy column names for data items like Data6 for Assets

- Comparison of the Legacy FTP Format to Xpressfeed
  - Excel files have old to new item mappings

WRDS Tour – When to ask for help?



Questions?

How to download data from WRDS?

- Five methods:
  - Web page
  - Python/Stata (PostgreSQL)
  - SFTP
  - SAS proc download
  - SAS studio

- Exercise 1
  - Download Fama-French times series factors

Download from WRDS - Web

- Fama-French factors
  - Daily or Monthly?
    - Let's do monthly
  - Choose date range
    - How to know date range available?
  - Tabs
    - Variable Description – often has a data dictionary and shows location of files
    - Manuals and Overview – has stuff, sometimes the actual manual (not this case)
    - Knowledgebase – additional resources that may be helpful
    - Data preview – allows you to see some of the data (very helpful to me)
      - Download and access data buttons new…

Download from WRDS - Web

- Fama-French factors example parameters
  - Date Range: 2000-1 to 2025-12
  - Mark all factor items
  - Output format
    - .csv
    - Uncompressed
    - MMDDYY10.
  - You can save query if you plan to use it several times
  - Press Submit Form button

Download from WRDS - Web

- How to view?
  - Press on page
  - Your account info

Browse Data

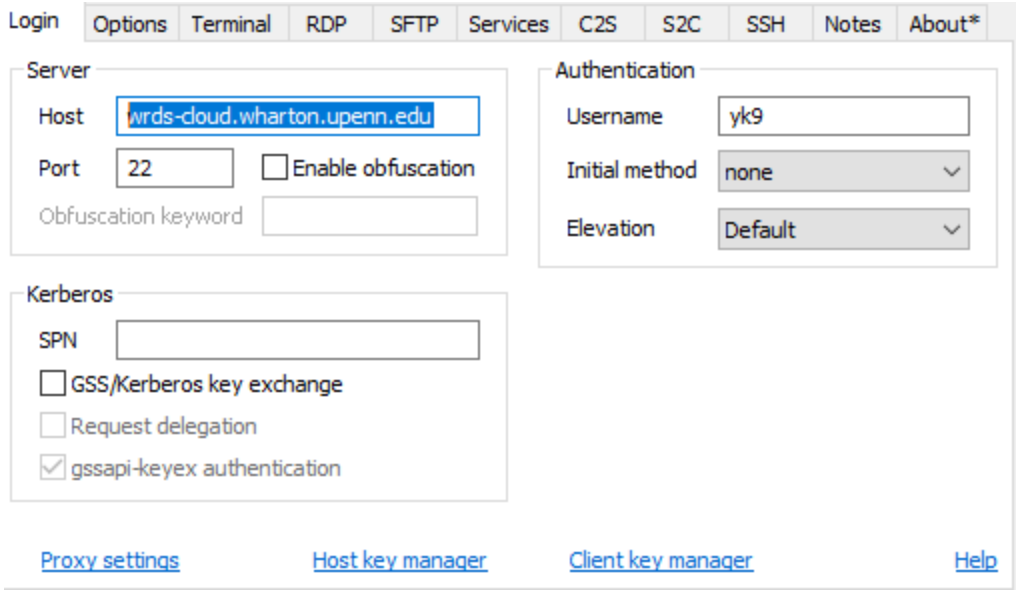
- Let’s say we want to see data for analyst forecast
  - https://wrds-www.wharton.upenn.edu/pages/get-data/ibes-thomson-reuters/ibes-academic/summary-history/summary-statistics/
    - Go to data preview
    - Select columns you want
      - E.g. Ticker, Statpers, Measure, fpi, estflag, numest, medest, meanest, stdev, fpedats, actual
    - Filters
      - Ticker = TSLA, statpers=leave default, measure=EPS, fpi=1, fpedats=leave default
      - Download the dataset
    - Press on download first 1k rows after we have what we want
      - You can construct mini samples to browse the data using this tool and download it

Download from WRDS - SFTP

- Use an SFTP software to connect to WRDS and transfer files
  - I like Bitvise (Windows Only)
    - https://www.bitvise.com/ssh-client-download
    - I have the 9.66 version
  - Some other people like Putty (not discussed)
    - https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html
  - WRDS recommendations (includes MacOS)
    - Access WRDS Remotely via SFTP (Secure FTP)
- You can use it to move code and data between your computer and the WRDS server

Download from WRDS - SFTP

- Install Bitvise (or any sftp app)
- Create a profile:



  - wrds-cloud.wharton.upenn.edu
  - Port: 22
  - Your WRDS username

Download from WRDS - SFTP

- Important places
  - Home Directory – 10GB space
    - /home/rice/username
  - Institution Directory – 500GB? Used as scratch drive
    - /scratch/rice
    - Gets deleted after 7? Days
  - Databases in SAS (prob not very useful anymore)
    - /wrdslin
- Storage info:
  - https://wrds-www.wharton.upenn.edu/pages/support/the-wrds-cloud/managing-data/storing-your-data-wrds-cloud/

Running batch jobs on WRDS Cloud

- Reference
  - https://wrds-www.wharton.upenn.edu/pages/support/the-wrds-cloud/running-jobs/batch-jobs-wrds-cloud/
- General info:
  - https://wrds-www.wharton.upenn.edu/pages/support/the-wrds-cloud/introduction-wrds-cloud/#jobs
- Max 5 jobs per user, 10 jobs per institution

Killing processes on WRDS

- Web and PC-SAS can be killed at “running queries”
- Batch jobs are killed by command line
  - qdel jobid (as shown on the web page for example)
  - qdel -u username (kills all batch job submitted for user)
    - Example: qdel -u user

- Python connections (Python session)
  - Harder or impossible to kill
  - Info
  - Querying (let’s review connection count, always always always close)

Questions?


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